ASCE 41-23 Tier 1 Checklists

FIRM:	
PROJECT NAME:	
SEISMICITY LEVEL:	
PROJECT NUMBER:	
COMPLETED BY:	
DATE COMPLETED:	
REVIEWED BY:	
REVIEW DATE:	

Legend: C = Compliant, NC = Noncompliant, N/A = Not Applicable, U = Unknown

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Project Name _____
Project Number _____

17.15CP Structural Checklist for Building Type PC2: Precast Concrete Frames with Shear Walls

Table 17-30. Collapse Prevention Structural Checklist for Building Type PC2.

Status				Evaluation Statement	Tier 2 Reference	Commentary Reference	Comments
Low and Moderate Seismicity							
Seismic-Force-Resisting System							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	CONCRETE BEARING WALLS: Floor and roof girders and trusses are not supported at the ends of concrete walls that are less than 10 in. (254 mm) thick. This statement only applies to framing supports located less than two times the wall thickness away from the wall end.	5.5.2.5.1	A.3.1.6.1	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	REDUNDANCY: The number of lines of shear walls in each principal direction is greater than or equal to 2.	5.5.1.1	A.3.2.1.1	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	SHEAR STRESS CHECK: The shear stress in the concrete shear walls, calculated using the Quick Check procedure of Section 4.4.3.3 is less than the greater of 100 lb/in.^{2} (0.69 MPa) or $2\sqrt{f'_c}$.	5.5.3.1.1	A.3.2.2.1	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	REINFORCING STEEL: The ratio of reinforcing steel area to gross concrete area is not less than 0.0012 in the vertical direction and 0.0020 in the horizontal direction.	5.5.3.1.3	A.3.2.2.2	
Diaphragms							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	TOPPING SLAB: Precast concrete diaphragm elements are interconnected by a continuous reinforced concrete topping slab with a minimum thickness of 2 in. (50.8 mm).	5.6.4	A.4.5.1	
Connections							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	TRANSFER TO SHEAR WALLS: Diaphragms are connected for transfer of seismic forces to the shear walls.	5.7.2	A.5.2.1	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	TOPPING SLAB TO WALLS OR FRAMES: Reinforced concrete topping slabs that interconnect the precast concrete diaphragm elements are doweled for transfer of forces into the shear wall or frame elements.	5.7.2	A.5.2.3	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	FOUNDATION DOWELS: Wall reinforcement is doweled into the foundation.	5.7.3.4	A.5.3.5	

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Status				Evaluation Statement	Tier 2 Reference	Commentary Reference	Comments
C ☐	NC ☐	N/A ☐	U ☐	GIRDER–COLUMN CONNECTION: There is a positive connection using plates, connection hardware, or straps between the girder and the column support.	5.7.4.1	A.5.4.1	
High Seismicity (Complete the Following Items in Addition to the Items for Low and Moderate Seismicity)							
Seismic-Force-Resisting System							
C ☐	NC ☐	N/A ☐	U ☐	PRECAST FRAMES: For buildings with concrete shear walls, precast concrete frame elements are not considered as primary components for resisting seismic forces.	5.5.2.4 5.5.2.5.1 5.5.2.5.2	A.3.1.5.2	
C ☐	NC ☐	N/A ☐	U ☐	PRECAST CONNECTIONS: For buildings with concrete shear walls, the connection between precast frame elements, such as chords, ties, and collectors in the seismic-force-resisting system, develops the capacity of the connected members.	5.6.1.1	A.3.1.5.3	
C ☐	NC ☐	N/A ☐	U ☐	DEFLECTION COMPATIBILITY: Secondary components have the shear capacity to develop the flexural strength of the components.	5.5.2.5.2	A.3.1.6.2	
C ☐	NC ☐	N/A ☐	U ☐	COUPLING BEAMS: Coupling beams have stirrups spaced at or less than d/2, and each wall or wall segment connected to the coupling beam is supported such that it can resist shear and overturning forces in the absence of the coupling beam. This statement only applies to coupling beams with span-to-depth ratios exceeding 2-to-1.	5.5.3.2.1	A.3.2.2.3	
Diaphragms							
C ☐	NC ☐	N/A ☐	U ☐	OPENINGS AT SHEAR WALLS: Diaphragm openings immediately adjacent to the shear walls are less than 25% of the wall length.	5.5.3.3.1	A.4.1.4	
Connections							
C ☐	NC ☐	N/A ☐	U ☐	UPLIFT AT PILE CAPS: Pile caps have top reinforcement, and piles are anchored to the pile caps.	5.7.3.5	A.5.3.8	
C ☐	NC ☐	N/A ☐	U ☐	CORBEL BEARING: If the frame girders bear on column corbels, the length of bearing is greater than 3 in. (76 mm).	5.7.4.3	A.5.4.3	
C ☐	NC ☐	N/A ☐	U ☐	CORBEL CONNECTIONS: The frame girders are not connected to corbels with welded elements.	5.7.4.3	A.5.4.4	

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